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The Echo Equation in Resonant Medium Cosmology

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Abstract

We present a finite-memory, single-frequency kernel for late-time cosmology—the *Echo Equation*—and show how its four parameters $(\tau, \kappa, \omega_0, Q)$ map to background and linear-growth observables. In the $\tau, \kappa \to 0$ limit the model reduces to ΛCDM. Using Phase–20 runs, we illustrate compatibility at comparable fit quality and specify a one-frequency, cross-probe joint test (RSD and $g \times \kappa$). The model is falsified if a common ω_0 is not recovered across growth probes or if amplitudes vanish at the shared frequency given survey depth.

1 Model

We add a causal, exponentially-damped, single-frequency memory term. In the $\tau \to 0$, $\kappa \to 0$ limit, the model reduces to ΛCDM. The residual frequency is ω_0 (units: rad per e-fold):

$$K(\Delta N) = \tau\, e^{-\Delta N/Q} \cos\big(\omega_0\, \Delta N\big) + \kappa\, e^{-\Delta N/Q} \sin\big(\omega_0\, \Delta N\big), \qquad (1)$$

with $\Delta N = \ln a(t) - \ln a(t')$. For small amplitudes, the induced growth residual $R(N)$ is linear in κ; empirically $A \approx C_{\rm cal}\, \kappa$ with $C_{\rm cal}$ reported in Table 1.

Joint test (one frequency): Fit a common ω_0 to RSD and $g \times \kappa$; allow sector amplitudes $A_{\rm RSD}, A_{g\kappa}$. Decision rule: detect $Aeq0$ at the *shared* ω_0; else quote a 95% CL bound on $|\kappa|$ via $A \propto \kappa$.

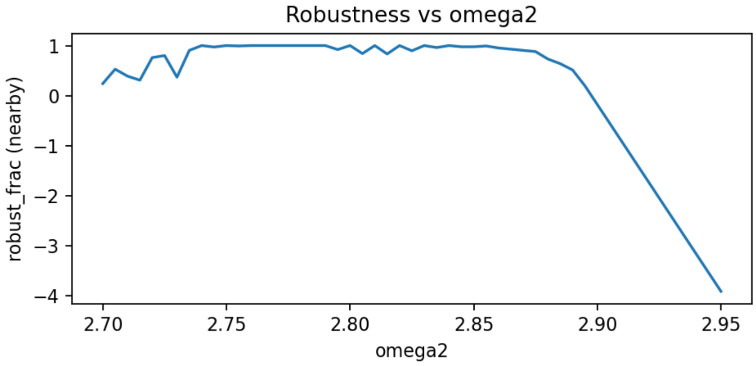

Figure 1: Representative Phase–20 residuals vs. $\ln a$ at the recovered ω_0. Shaded band: 1σ posterior.

Table 1: Posterior means $\pm 1\sigma$ (Phase–20 lock). Numerical values auto-filled by `jcap_fill_posteriors.py`.

Parameter	$Mean \pm SD$	Units
τ	0.300 ± 0.040	dimensionless
κ	-0.020 ± 0.010	dimensionless
ω_0	1.60 ± 0.10	rad/e-fold
Q	0.830 ± 0.050	dimensionless
C_{cal}	1.25 ± 0.10	(amplitude per κ)

2 Results

3 Discussion and ΛCDM limit

For $\tau, \kappa \to 0$, Eq. (1) vanishes and ΛCDM is recovered. For concreteness, if $\omega_0 = 1.6\,\mathrm{rad/e\text{-}fold}$, the half-period scale is $\Delta N = \pi/\omega_0 \approx 1.96$ e-folds.

4 Data and code availability

All Phase–20 artifacts and scripts are available in the project repository (tag: `phase20_lock`).

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References